

Multiple Choice (10 marks)

1. What process in the plant is responsible for the phenomenon of guttation?
 - (a) Guttation is the outcome of rapid cellular respiration in plant tissues.
 - (b) Guttation is a result of photosynthesis.
 - (c) Guttation is caused by external weather conditions.
 - (d) Guttation is explained by the effect of root pressure.
 - (e) Guttation is a byproduct of the plant's defense mechanisms.
2. Chromosomal sex in mammals has been shown to be strongly Y chromosome determined. How was this shown to be true?
 - (a) The absence of a Y chromosome determines maleness in mammals
 - (b) The presence of a Y chromosome usually determines maleness in mammals
 - (c) The ratio of two X chromosomes to one Y chromosome determines maleness
 - (d) The presence of an X chromosome determines maleness in mammals
 - (e) The presence of an X chromosome in the absence of a Y chromosome determines maleness
3. What prevents the stomach from being digested by its own secretions?
 - (a) The stomach has a high permeability to hydrogen ions
 - (b) The presence of beneficial bacteria that neutralize stomach acid
 - (c) The stomach lining is replaced every month
 - (d) The stomach lining secretes digestive enzymes only when food is present
 - (e) The production of bile in the stomach neutralizes the digestive acids
4. The acrosome and cortical reactions that occur immediately after fertilization ensure which of the following?
 - (a) The zygote will not divide into two embryos, producing twins.
 - (b) The sperm will not be able to fertilize the egg.
 - (c) The sperm will fertilize the ovum.
 - (d) The sperm will fertilize multiple eggs.
 - (e) The zygote will begin to divide normally.
5. Black-headed gulls remove broken eggshells from their nests immediately after the young have hatched. Explain the biological significance of this behavior.

- (a) It attracts predators away from the nestlings.
- (b) It aids in temperature regulation of the nest by reducing insulation.
- (c) It assists in the hatching of other eggs.
- (d) It is a ritualistic behavior without any particular significance.
- (e) It signals to other gulls that the nesting site is occupied.

True / False (10 marks)

Answer True or False

- 6. True or False: Is routine dissection of the station 9 lymph nodes really necessary for primary lung cancer?
- 7. True or False: Does the lipid-lowering peroxisome proliferator-activated receptors ligand bezafibrate prevent colon cancer in patients with coronary artery disease?
- 8. True or False: Target Serum Urate: Do Gout Patients Know Their Goal?
- 9. True or False: Attaining negative margins in breast-conservation operations: is there a consensus among breast surgeons?
- 10. True or False: Does a family meetings intervention prevent depression and anxiety in family caregivers of dementia patients?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the mechanisms through which a newborn can contract syphilis from an infected mother, including relevant stages of pregnancy and birth.
- 12. Discuss the physiological mechanisms by which reduced salt intake can impact blood pressure in hypertensive patients, including the role of blood volume and water retention.

End of Paper